

Association Rule Learning: Market Basket Analysis of the Groceries Dataset

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Course: Unsupervised Learning Methods
Assignment: Association Rule Learning
Lecturer: Dr. Pacifique Nizeyimana
Dataset: Groceries
Software: Python (pandas, mlxtend)

Question 1: Understand the Data

The Groceries dataset was loaded as a list of transactions, where each transaction is the list of items purchased together in a single shopping trip. The raw CSV file has no header row and a variable number of items per row, so each row was read directly and stripped of empty entries to reconstruct the true item list per transaction.

- a. **Number of transactions:** 9,835
- b. **Number of unique items:** 169
- c. **Ten most frequently purchased items:** see Table 1.

Table 1: Ten most frequently purchased items.

Item	Count
whole milk	2,513
other vegetables	1,903
rolls/buns	1,809
soda	1,715
yogurt	1,372
bottled water	1,087
root vegetables	1,072
tropical fruit	1,032
shopping bags	969
sausage	924

Whole milk is the clear leader, appearing in more than a quarter of all transactions. Dairy and produce dominate the top ten, consistent with weekly grocery shopping staples.

Question 2: Frequent Itemsets

Frequent itemsets were identified using the Apriori algorithm (mlxtend), with a minimum support threshold of 0.01 (1%). This threshold was chosen deliberately low: with 169 distinct items across 9,835 transactions, most item combinations naturally co-occur in only a small fraction of transactions, so a higher threshold returns very few or zero itemsets. This search returned 245 frequent itemsets containing two or more items.

- a. **At least 10 frequent itemsets (2+ items):** see Table 2.
- b. **Support values:** reported directly in Table 2.
- c. **Highest-support itemset:** {other vegetables, whole milk}, support = 0.0748.
- d. **Interpretation of support:** support measures how common a combination is across all shopping trips — a raw popularity measure, not a relationship measure. The highest-support itemsets here are dominated by the two individually most popular items (whole milk and other vegetables), which mechanically inflates their pairwise support regardless of whether there is any special connection between them. Support alone therefore tells us which combinations are common, but not whether that commonality is meaningful or coincidental — that distinction requires confidence and lift (Question 3).

Table 2: Ten frequent itemsets (2+ items) with the highest support.

Itemset	Support
{other vegetables, whole milk}	0.0748
{rolls/buns, whole milk}	0.0566
{yogurt, whole milk}	0.0560
{root vegetables, whole milk}	0.0489
{root vegetables, other vegetables}	0.0474
{yogurt, other vegetables}	0.0434
{rolls/buns, other vegetables}	0.0426
{tropical fruit, whole milk}	0.0423
{soda, whole milk}	0.0401
{rolls/buns, soda}	0.0383

Question 3: Association Rules

Association rules were generated from the frequent itemsets using mlxtend’s `association_rules()` function, with a minimum confidence threshold of 0.3. This produced 125 rules in total; the ten with the highest confidence are reported in Table 3.

Question 4: Understanding the Measures

Three rules were selected to illustrate the three measures:

Rule 1: {root vegetables, citrus fruit} $\rightarrow$ {other vegetables} — Support = 0.0104, Confidence = 0.586, Lift = 3.03.

Table 3: Ten association rules with the highest confidence.

Antecedent	Consequent	Support	Confidence	Lift
{citrus fruit, root vegetables}	{other vegetables}	0.0104	0.586	3.030
{root vegetables, tropical fruit}	{other vegetables}	0.0123	0.585	3.021
{curd, yogurt}	{whole milk}	0.0101	0.582	2.279
{butter, other vegetables}	{whole milk}	0.0115	0.574	2.245
{root vegetables, tropical fruit}	{whole milk}	0.0120	0.570	2.231
{root vegetables, yogurt}	{whole milk}	0.0145	0.563	2.203
{domestic eggs, other vegetables}	{whole milk}	0.0123	0.553	2.162
{whipped/sour cream, yogurt}	{whole milk}	0.0109	0.525	2.053
{root vegetables, rolls/buns}	{whole milk}	0.0127	0.523	2.047
{other vegetables, pip fruit}	{whole milk}	0.0135	0.518	2.025

Rule 2: {curd, yogurt} $\rightarrow$ {whole milk} — Support = 0.0101, Confidence = 0.582, Lift = 2.28.

Rule 3: {other vegetables} $\rightarrow$ {whole milk} — Support = 0.0748, Confidence = 0.387, Lift = 1.51.

(a) **Support** measures how often the entire itemset (antecedent and consequent together) appears across all transactions. Rule 3’s support of 0.0748 means these two items were bought together in 7.48% of all 9,835 transactions — the highest of the three, since both items are individually very common.

(b) **Confidence** measures, out of all transactions containing the antecedent, what fraction also contain the consequent. Rule 1’s confidence of 0.586 means that among customers who bought both root vegetables and citrus fruit, 58.6% of them also bought other vegetables.

(c) **Lift** measures how much more likely the consequent is, given the antecedent, compared to its normal likelihood on its own. Rule 1’s lift of 3.03 means customers who buy root vegetables and citrus fruit are 3.03 times more likely to also buy other vegetables than a random customer. Rule 3’s much lower lift (1.51) shows that despite having by far the highest support, it reflects a much weaker genuine relationship: whole milk is so universally popular (bought in 25.5% of all transactions) that almost anything shows moderately high co-occurrence with it, without a strong real connection.

Question 5: Find Interesting Rules

- Highest confidence:** {citrus fruit, root vegetables} $\rightarrow$ {other vegetables}, confidence = 0.586.
- Highest lift:** {citrus fruit, other vegetables} $\rightarrow$ {root vegetables}, lift = 3.295.
- Highest support:** {other vegetables} $\rightarrow$ {whole milk}, support = 0.0748.
- Are these the same rule?** No — all three are different rules. This demonstrates that support, confidence, and lift each measure something distinct: support rewards raw popularity, confidence rewards a strong one-directional purchase pattern, and lift rewards a genuinely surprising relationship relative to how common the consequent already is on its own. The highest-support rule simply involves two extremely common items, while the highest-lift rule

involves a smaller, more specific combination that is far more strongly connected than chance alone would predict.

Question 6: Interpret a Rule

Selected rule: {root vegetables, citrus fruit} $\rightarrow$ {other vegetables}, Support = 0.0104, Confidence = 0.586, Lift = 3.03.

In simple language: Customers who purchase both root vegetables and citrus fruit also tend to purchase other vegetables, in about 59% of their shopping trips.

Strong or weak association? This is a **strong** association. A lift of 3.03 means this combination occurs three times more often than would be expected if the two purchases were independent of one another — a lift this far above 1 indicates a genuine, non-random purchasing pattern (likely reflecting a general “healthy/produce-focused” shopping trip), rather than coincidence.

Question 7: Business Application

(a) **Two products to place close together:** Root vegetables and other vegetables — their pairwise rule shows a strong lift, and placing them on adjacent shelving would match how customers already shop.

(b) **Possible product bundle:** A “vegetable basket” bundle combining root vegetables, citrus fruit, and other vegetables, directly reflecting the highest-confidence rule found in Question 5, where 58.6% of customers who bought the first two items also bought the third.

(c) **Recommendation system example:** When a customer’s basket already contains root vegetables and citrus fruit, the store’s recommendation system (whether an in-store app notification or an online checkout prompt) could automatically suggest other vegetables, since historical data shows a 58.6% chance the customer will want them too — directly operationalizing the rule discovered in this analysis.

Question 8: Understanding Association

(a) **Does an association rule prove causation?** No. An association rule only shows that two items were frequently purchased together in the historical data; it says nothing about *why*. Buying root vegetables does not cause someone to buy other vegetables — both purchases more likely stem from a shared underlying cause, such as a general preference for vegetable-heavy meals. This is the same principle as “correlation does not imply causation.”

(b) **Why might a rule have high confidence but low lift?** This happens when the consequent item is already extremely popular on its own, independent of the antecedent. For example, the rule {berries} $\rightarrow$ {whole milk} has a confidence of 0.355 (35.5% of berry-buyers also buy milk), but its lift is only 1.39 — barely above 1. This is because whole milk alone already appears in 25.5% of all transactions, so a 35.5% co-occurrence rate is not actually much higher than what random chance alone would produce. High confidence can therefore be misleading if the consequent is simply a universally popular item.

(c) **What does a lift value greater than 1 indicate?** A lift greater than 1 means the two items are purchased together more often than would be expected if they were completely independent — a positive, genuine association. A lift of exactly 1 would mean no relationship at all (pure chance), and a lift below 1 would mean the items are actually purchased together *less* than chance would predict (a negative association).